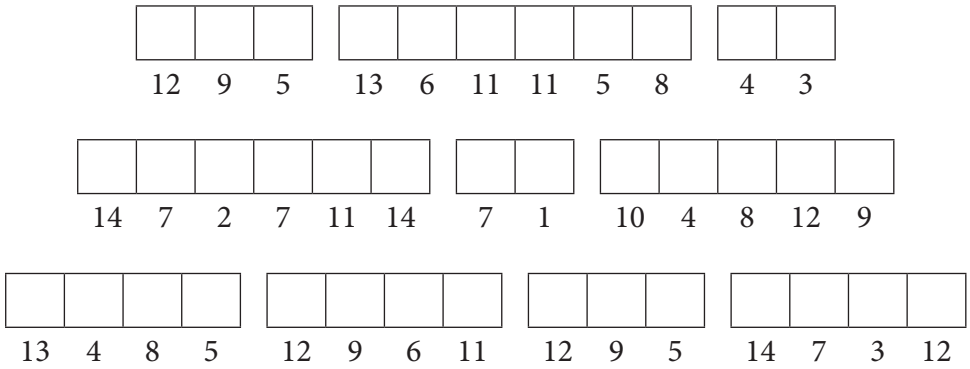
Name: _____

WORKSHEET

Complex conjugates and inverses



Pierre Corneille

Questions

What is the complex conjugate of:

1 $3 - 5i$

2 $7 + 4i$

3 $-2i - 9$

What is the additive inverse of:

4 $6 + 13i$

5 $1 - 5i$

6 $-4 + 7i$

What is the multiplicative inverse of:

7 $\dfrac{1}{8 - i}$

8 $9 + 4i$

9 $2 - 7i$

Express the following in the form $a + bi$:

10 $\dfrac{1}{1+i}$

11 $\dfrac{5i}{3 - 4i}$

12 $\dfrac{2 - 7i}{2 + 7i}$

13 $\dfrac{8 + 11i}{4 - 3i}$

14 $\dfrac{2i}{5 - 6i} + \dfrac{3 - 12i}{7 + 4i}$

Solutions

A $4 - 7i$

E $-1 + 5i$

F $-9 + 2i$

G $-\dfrac{2427}{3965} - \dfrac{5206}{3965}i$

H $\dfrac{2}{53} + \dfrac{7}{53}i$

I $8 - i$

M $-\dfrac{1}{25} + \dfrac{68}{25}i$

N $-\dfrac{4}{5} + \dfrac{3}{5}i$

O $-6 - 13i$

R $\dfrac{9}{97} - \dfrac{4}{97}i$

S $3 + 5i$

T $-\dfrac{45}{53} - \dfrac{28}{53}i$

V $7 - 4i$

W $\dfrac{1}{2} - \dfrac{1}{2}i$

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